```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import files
uploaded=files.upload()
print(uploaded)
```

Choose Files train.csv

train.csv(text/csv) - 2129689 bytes, last modified: 9/11/2020 - 100% done

```
Saving train.csv to train.csv
{'train.csv': b'Row ID,Order ID,Order Date,Ship Date,Ship Mode,Customer ID,Customer Name,Segment,Country,City,State,P
```

```
df = pd.read_csv('train.csv')
```

```
df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Pr
0	1	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	Fu 10
1	2	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	Fu 10
2	3	CA-2017-138688	12/06/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036.0	West	C 10
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	F 10
4	5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	C 10

Next steps: Generate code with df New interactive sheet

```
df['Order Date'] = pd.to_datetime(df['Order Date'], dayfirst=True)
```

```
df['Ship Date'] = pd.to_datetime(df['Ship Date'], dayfirst=True)
```

```
df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID
0	1	CA-2017-152156	2017-11-08	2017-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-BO-10001798
1	2	CA-2017-152156	2017-11-08	2017-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-CH-10000454
2	3	CA-2017-138688	2017-06-12	2017-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036.0	West	OFF-LA-10000240
3	4	US-2016-108966	2016-10-11	2016-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	FUR-TA-10000577
4	5	US-2016-108966	2016-10-11	2016-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	OFF-ST-10000760

Next steps: Generate code with df New interactive sheet

```
df.isnull().sum()
```

	0
Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Country	0
City	0
State	0
Postal Code	11
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0

dtype: int64

```
df['Postal Code'] = df['Postal Code'].fillna("Unknown")
```

```
df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID
0	1	CA-2017-152156	2017-11-08	2017-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-BO-10001798
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4	5	US-2016-108966	2016-10-11	2016-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	OFF-ST-10000760

Next steps: Generate code with df New interactive sheet

```
df['delivery_days'] = (df['Ship Date'] - df['Order Date']).dt.days
```

```
df['delivery_status'] = df['delivery_days'].apply(lambda x: "On Time" if x <= 5 else "Late")
```

```
def speed(x):
    if x <= 2:
        return "Fast"
    elif x <= 5:
        return "Normal"
    else:
        return "Slow"

df['delivery_speed'] = df['delivery_days'].apply(speed)
```

```
df['year'] = df['Order Date'].dt.year
df['month'] = df['Order Date'].dt.month
```

```
df[['delivery_days','delivery_status','delivery_speed']].head()
```

	delivery_days	delivery_status	delivery_speed
0	3	On Time	Normal
1	3	On Time	Normal
2	4	On Time	Normal
3	7	Late	Slow
4	7	Late	Slow

```
df.groupby('Ship Mode')['delivery_days'].mean()
```

Ship Mode	delivery_days
First Class	2.179214
Same Day	0.044610
Second Class	3.249211
Standard Class	5.008363

dtype: float64

```
df['delivery_status'].value_counts(normalize=True) * 100
```

delivery_status	proportion
On Time	81.785714
Late	18.214286

dtype: float64

```
df.groupby('Region')['Sales'].sum().sort_values(ascending=False)
```

Region	Sales
West	710219.6845
East	669518.7260
Central	492646.9132
South	389151.4590

dtype: float64

```
df.columns
```

```
Index(['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode',
       'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State',
       'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category',
       'Product Name', 'Sales', 'delivery_days', 'delivery_status',
       'delivery_speed', 'year', 'month'],
      dtype='object')
```

```
df.columns = df.columns.str.lower().str.replace(" ", "_").str.replace("-", "_")
```

```
df.head()
```

	row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country	city	...	pro
0	1	CA-2017-152156	2017-11-08	2017-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	1